

Target SQL Business Case

Introduction:

Target is a leading global retail brand known for providing high-quality products and an exceptional customer experience. This case study analyzes Target's e-commerce operations in Brazil using data from approximately 100,000 orders placed between 2016 and 2018. The dataset includes information on orders, customers, products, payments, deliveries, and reviews, enabling a comprehensive assessment of business performance and customer behaviour.

Objective:

The objective of this analysis is to understand customer purchasing behavior and order patterns across different states in Brazil. The study examines key metrics such as average product cost, delivery time, payment methods, and freight charges to identify trends and uncover opportunities for improving logistics efficiency, delivery performance, and overall customer experience.

Exploratory Data Analysis (EDA)

The dataset consists of eight CSV files containing information on customers, geolocations, orders, order items, payments, reviews, products, and sellers.

Dataset Schema

Table Name -Description

Customers	- Customer information and location details
Geolocation	- Geographic coordinates and zip code information
Orders	- Order lifecycle and delivery details
order_items	- Product-level details for each order
Payments	- Payment methods and transaction values
Reviews	- Customer ratings and reviews
Products	- Product information and categories
Sellers	- Seller information and locations

Orders Table Schema:

Query:

select

COLUMN_NAME, DATA_TYPE

from ecommerce-406417.Target.INFORMATION_SCHEMA.COLUMNS

where table_name = 'orders'

Row	COLUMN_NAME ▾	DATA_TYPE ▾
1	order_id	STRING
2	customer_id	STRING
3	order_status	STRING
4	order_purchase_timestamp	TIMESTAMP
5	order_approved_at	TIMESTAMP
6	order_delivered_carrier_date	TIMESTAMP
7	order_delivered_customer_date	TIMESTAMP
8	order_estimated_delivery_date	TIMESTAMP

Order Time Range:

The time range between the orders were placed

```
select min(order_purchase_timestamp) as first_order,  
       max(order_purchase_timestamp) as last_order  
from `Target.orders`
```

Row	first_order ▾	last_order ▾
1	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC

- The dataset contains orders placed between **September 4, 2016** and **October 17, 2018**.

Number of unique customers

```
select count(distinct customer_unique_id) as coustomer_count  
from `Target.customers`;
```

Row	coustomer_count ▾
1	96096

- **96,096 unique customers** made purchases between 2016 and 2018.

The Geographic Coverage of Customers Across States And Cities In Brazil

```
select count(distinct customer_state) state_count,  
       count(distinct customer_city) city_count  
from `Target.customers`
```

Row	state_count	city_count
1	27	4119

Target's customers are distributed across all 27 Brazilian states and 4,119 cities, indicating a broad nationwide presence and customer reach.

The distribution of customers across Brazilian states:

Top States by Number of Customers

```
select customer_state,  
       count(customer_unique_id) as customer_count  
from `Target.customers`  
group by 1  
order by 2 desc;
```

Row	customer_state	customer_count
1	SP	41746
2	RJ	12852
3	MG	11635
4	RS	5466
5	PR	5045
6	SC	3637
7	BA	3380
8	DF	2140
9	ES	2033
10	GO	2020
11	PE	1652
12	CE	1336
13	PA	975

SP has the largest customer base with 41,746 customers, significantly outperforming all other states, followed by RJ and MG, indicating a strong concentration of customers in Brazil's southeastern region.

The Cities With the Highest Number of Customers:

```
select customer_city, count(customer_unique_id) as customer_count  
from `Target.customers`  
group by 1
```

order by 2 desc;

Row	customer_city	customer_count
1	sao paulo	15540
2	rio de janeiro	6882
3	belo horizonte	2773
4	brasilia	2131
5	curitiba	1521
6	campinas	1444
7	porto alegre	1379
8	salvador	1245
9	guarulhos	1189
10	sao bernardo do campo	938

São Paulo leads with 15,540 customers, followed by Rio de Janeiro (6,882) and Belo Horizonte (2,773), indicating that a significant share of customers is concentrated in a few major urban centers.

The Yearly Distribution of Orders:

```
select year ,
       count(order_id) as order_count
from
       (select extract(year from order_purchase_timestamp) `year`, order_id
        from `Target.orders`)t
group by 1
order by 1;
```

Row	year	order_count
1	2016	329
2	2017	45101
3	2018	54011

Order volume increased significantly from 45,101 orders in 2017 to 54,011 orders in 2018, while 2016 recorded only 329 orders due to limited data availability for that year.

Year-Over-Year Order Growth:

```
with order_per_year as
(select year ,
       count(order_id) as order_count
from
       (select extract(year from order_purchase_timestamp) `year`, order_id
        from `Target.orders`)t
group by 1
```

```

order by 1
)
select year,
       order_count,
       lag(order_count) over (order by year) as prev_order_count,
       round(100*(order_count - lag(order_count) over (order by year))/lag(order_count) over
            (order by year),2) as growth_percent
from order_per_year;

```

Row	year	order_count	prev_order_count	growth_percent
1	2016	329	null	null
2	2017	45101	329	13608.51
3	2018	54011	45101	19.76

Order volume grew by 19.76% in 2018 compared to 2017, indicating continued growth in customer demand and business activity.

Monthly Order Trends Across Years:

Seasonality

```

with order_per_month as
(
select extract(year from order_purchase_timestamp) `year`,
       extract(month from order_purchase_timestamp) `month`,
       count(order_id) as order_count
from `Target.orders`
group by 1,2
order by 1,2
)
select month,
       sum(case when year=2016 then order_count else 0 end) as y_2016,
       sum(case when year=2017 then order_count else 0 end) as y_2017,
       sum(case when year=2018 then order_count else 0 end) as y_2018
from order_per_month
group by 1
order by 1;

```

Row	month	y_2016	y_2017	y_2018
1	1	0	800	7269
2	2	0	1780	6728
3	3	0	2682	7211
4	4	0	2404	6939
5	5	0	3700	6873
6	6	0	3245	6167
7	7	0	4026	6292
8	8	0	4331	6512
9	9	4	4285	16
10	10	324	4631	4

Order volume generally increased throughout the period, with a notable surge observed from November 2017 through March 2018, indicating increase in customer demand and business activity.

Year-Over-Year Order Trends Across States

Create View `ecommerce-406417`.`Target`.`order\_details` as

```
select c.customer_state,
       extract(year from o.order_purchase_timestamp) as `year`,
       extract(month from o.order_purchase_timestamp) as `month`,
       count(o.customer_id) as `order_count`
from `Target.orders` o inner join `Target.customers` c
on o.customer_id=c.customer_id
group by c.customer_state,year,month
order by year,month;
```

```
select customer_state,
       sum(case when year=2016 then order_count else 0 end) as y_2016,
       sum(case when year=2017 then order_count else 0 end) as y_2017,
       sum(case when year=2018 then order_count else 0 end) as y_2018
from `Target.order_details`
group by customer_state
order by 2 desc,3,4;
```

Row	customer_state	y_2016	y_2017	y_2018
1	SP	115	17760	23871
2	RJ	56	6225	6571
3	MG	40	5414	6181
4	RS	25	2661	2780
5	PR	20	2270	2755
6	SC	11	1720	1906
7	GO	9	955	1056
8	CE	8	660	668
9	PE	7	772	873
10	DF	6	921	1213

São Paulo (SP) consistently recorded the highest order volume and showed substantial growth from 17,760 orders in 2017 to 23,871 in 2018, reinforcing its position as the primary market. Most states also experienced growth in 2018, indicating expanding demand across regions.

The Time of Day When Customers Place Orders:

```
with `cte` as
(
select order_time,
       count(order_id) as `order_count`
from
  (select
    order_id,extract(year from order_purchase_timestamp) as `year`,extract(month from
order_purchase_timestamp) as `month`,
    case when extract(time from order_purchase_timestamp) between ("00:00:00") and
("06:59:59") then "Dawn"
    when extract(time from order_purchase_timestamp) between ("07:00:00") and
("12:59:59") then "Morning"
    when extract(time from order_purchase_timestamp) between ("13:00:00") and
("18:59:59") then "afternoon"
    when extract(time from order_purchase_timestamp) between ("19:00:00") and
("23:59:59") then "night"
    end as `order_time`
  from Target.orders
)
group by order_time
order by order_time
)
select order_time,
       order_count
from cte
order by order_count desc;
```

Row	order_time	order_count
1	afternoon	38135
2	night	28331
3	Morning	27733
4	Dawn	5242

Order activity peaks in the afternoon and remains strong during the evening, while dawn records the lowest number of orders.

Revenue Growth Between 2017 and 2018

```
with order_amount as
(
```

```

select extract(year from o.order_purchase_timestamp) order_year,
       extract(month from o.order_purchase_timestamp) order_month,
       round(sum(p.payment_value),2) as order_value
from `Target.orders` o inner join `Target.payments` p
on o.order_id = p.order_id
group by 1,2
order by 1,2
),
per_year_order_amount as
(
  select order_month,
         sum(case when order_year=2017 then order_value else 0 end) as order_value_17,
         sum(case when order_year=2018 then order_value else 0 end) as order_value_18
  from order_amount
  where order_month between 0 and 8
  group by 1
  order by 1
)
select order_month,
       order_value_17,
       order_value_18,
       round((((order_value_18-order_value_17)/order_value_17)*100,2) as
       order_value_growth_percentage
from per_year_order_amount

```

Row	order_month	order_value_17	order_value_18	order_value_grow...
1	1	138488.04	1115004.18	705.13
2	2	291908.01	992463.34	239.99
3	3	449863.6	1159652.12	157.78
4	4	417788.03	1160785.48	177.84
5	5	592918.82	1153982.15	94.63
6	6	511276.38	1023880.5	100.26
7	7	592382.92	1066540.75	80.04
8	8	674396.32	1022425.32	51.61

Order value increased across all comparable months in 2018, with the highest growth observed in January (705.13%), indicating a substantial increase in sales value compared to the previous year.

Total and Average Revenue Across States

Top 10 states

```

select c.customer_state,
       round(sum(pm.payment_value),2) as total_revenue,
       round(avg(pm.payment_value),2) as avg_revenue
from `Target.orders` o inner join `Target.customers` c

```



```

on o.customer_id= c.customer_id
inner join `Target.payments` pm
on o.order_id= pm.order_id
group by 1
order by 2 desc
limit 10;

```

Row	customer_state	total_revenue	avg_revenue
1	SP	5998226.96	137.5
2	RJ	2144379.69	158.53
3	MG	1872257.26	154.71
4	RS	890898.54	157.18
5	PR	811156.38	154.15
6	SC	623086.43	165.98
7	BA	616645.82	170.82
8	DF	355141.08	161.13
9	GO	350092.31	165.76
10	ES	325967.55	154.71

São Paulo (SP) generated the highest revenue by a significant margin, followed by Rio de Janeiro (RJ) and Minas Gerais (MG), highlighting these states as the primary contributors to overall revenue. Average revenue per order remains relatively consistent across the top-performing states.

Lowest 10 States

```

select c.customer_state,
       round(sum(pm.payment_value),2) as total_revenue,
       round(avg(pm.payment_value),2) as avg_revenue
from `Target.orders` o inner join `Target.customers` c
on o.customer_id= c.customer_id
inner join `Target.payments` pm
on o.order_id= pm.order_id
group by 1
order by 2 asc
limit 10;

```

Row	customer_state ▼	total_revenue ▼	avg_revenue ▼
1	RR	10064.62	218.8
2	AP	16262.8	232.33
3	AC	19680.62	234.29
4	AM	27966.93	181.6
5	RO	60866.2	233.2
6	TO	61485.33	204.27
7	SE	75246.25	208.44
8	AL	96962.06	227.08
9	RN	102718.13	196.78
10	PI	108523.97	207.11

Roraima (RR), Amapá (AP), and Acre (AC) generated the lowest total revenue, indicating limited business activity in these states. However, their average revenue per order is relatively high, suggesting that lower revenue is driven by fewer orders rather than lower spending per order.

Total & Average Order Price Across States

```
select c.customer_state,
       round(sum(price),2) as `total_order_price`,
       round(avg(price),2) as `avg_order_price`
from `Target.orders` o inner join `Target.customers` c
on o.customer_id=c.customer_id
inner join `Target.order_items` oi
on o.order_id=oi.order_id
group by c.customer_state
order by c.customer_state;
```

Row	customer_state ▼	total_order_price ▼	avg_order_price ▼
1	AC	15982.95	173.73
2	AL	80314.81	180.89
3	AM	22356.84	135.5
4	AP	13474.3	164.32
5	BA	511349.99	134.6
6	CE	227254.71	153.76
7	DF	302603.94	125.77
8	ES	275037.31	121.91
9	GO	294591.95	126.27
10	MA	119648.22	145.2
11	MG	1585308.03	120.75
12	MS	116812.64	142.63
13	MT	156453.53	148.3

SP, RJ, and MG contribute the most to total order value, while average order prices remain relatively consistent, generally ranging between 120 and 180, indicating similar spending behavior across regions.

Total & Average Freight Charges Across States

```
select c.customer_state,
       round(sum(freight_value),2) as `total_order_price`,
       round(avg(freight_value),2) as `avg_order_price`
from `Target.orders` o inner join `Target.customers` c
on o.customer_id=c.customer_id
inner join `Target.order_items` oi
on o.order_id=oi.order_id
group by c.customer_state
order by c.customer_state;
```

Row	customer_state	total_freight_value	avg_freight_value
1	AC	3686.75	40.07
2	AL	15914.59	35.84
3	AM	5478.89	33.21
4	AP	2788.5	34.01
5	BA	100156.68	26.36
6	CE	48351.59	32.71
7	DF	50625.5	21.04
8	ES	49764.6	22.06
9	GO	53114.98	22.77
10	MA	31523.77	38.26

Average freight charges range from 15.15 in São Paulo (SP) to 42.98 in Roraima (RR), indicating substantial regional variation in shipping costs. States in the North and Northeast generally incur higher freight charges, while South and Southeast states tend to have lower shipping costs, suggesting differences in logistics accessibility and delivery distances.

States With the Highest and Lowest Average Delivery Times

```
with `cte` as
(
select customer_state,
       round(avg(date_diff(order_delivered_customer_date, order_purchase_timestamp, day)))
as `avg_delivery_time`,
       dense_rank()over(order by avg(date_diff(order_delivered_customer_date,
order_purchase_timestamp,day)) desc ) as `rnk_desc`,
       dense_rank()over(order by avg(date_diff(order_delivered_customer_date,
order_purchase_timestamp,day))asc) as `rnk_asc`
from `Target.orders` o inner join `Target.customers` c
on o.customer_id=c.customer_id
```

```

group by customer_state
)
select c1.rnk_desc,
       c1.customer_state,
       c1.avg_delivery_time as `highest_avg_delivery_time`,
       c2.customer_state,
       c2.avg_delivery_time as `lowest_avg_delivery_time`
from cte c1 inner join cte c2
on c1.rnk_desc=c2.rnk_asc
where c1.rnk_desc<=5
order by c1.rnk_desc, c2.rnk_asc;

```

Row	rnk_desc	customer_state	highest_avg_delive...	customer_state_1	lowest_avg_deliv...
1	1	RR	29.0	SP	8.0
2	2	AP	27.0	PR	12.0
3	3	AM	26.0	MG	12.0
4	4	AL	24.0	DF	13.0
5	5	PA	23.0	SC	14.0

Delivery times vary significantly across states, with Roraima (RR), Amapá (AP), and Amazonas (AM) experiencing the longest average delivery times of 26–29 days, while São Paulo (SP), Paraná (PR), and Minas Gerais (MG) receive orders in as little as 8–12 days, indicating more efficient logistics in these regions.

The States Where Orders Are Delivered Significantly Earlier Than The Estimated Delivery Date.

```

with order_delivery_time as
(
select customer_state,
       date_diff(order_delivered_customer_date,order_purchase_timestamp, day) as `
       actual_delivery_time`,
       date_diff(order_estimated_delivery_date,order_purchase_timestamp,day) as `
       estimated_delivery_time`
from `Target.orders` o inner join `Target.customers` c
on o.customer_id=c.customer_id
where order_status="delivered"
)
select customer_state,
       round(avg(estimated_delivery_time),2) as `avg_estimated_delivery_time`,
       round(avg(estimated_delivery_time - actual_delivery_time),2) as
       `avg_actual_delivery_time`
from order_delivery_time
group by customer_state
order by `avg_actual_delivery_time` desc
limit 10

```

;

Row	customer_state	avg_estimated_d...	avg_actual_delive...
1	AC	40.72	20.09
2	RO	38.39	19.47
3	AP	45.87	19.13
4	AM	44.92	18.94
5	RR	45.63	16.66
6	MT	31.37	13.77
7	PA	36.79	13.48
8	RS	28.16	13.34
9	RN	31.87	13.05
10	PR	24.25	12.73

Orders in AC, RO, and AP are delivered nearly 20 days earlier than the estimated delivery date, suggesting substantial room to improve delivery time predictions.

Top-Performing Product Categories Based on Order Volume and Revenue Generation

```
select p.product_category,
       count(o.order_id) as order_count,
       round(sum(payment_value),2) as order_value
from `Target.orders` o
inner join `Target.payments` pm
on o.order_id= pm.order_id
inner join `Target.order_items` oi
on o.order_id= oi.order_id
inner join `Target.products` p
on oi.product_id= p.product_id
group by 1
order by 2 desc
limit 10;
```

Row	product_category	order_count	order_value
1	bed table bath	11823	1712553.67
2	HEALTH BEAUTY	9972	1657373.12
3	sport leisure	8945	1392127.56
4	Furniture Decoration	8744	1430176.39
5	computer accessories	8082	1585330.45
6	housewares	7355	1094758.13
7	Watches present	6201	1429216.68
8	telephony	4721	486882.05
9	Garden tools	4574	838280.75
10	automotive	4379	852294.33

Home, technology, and personal care categories dominate both order volume and revenue, with 'Bed Table Bath' emerging as the most frequently purchased product category.

The Product Categories Receiving The Highest Number Of Customer Reviews

```
select
  p.product_category,
  count(r.review_id) as total_reviews
from `Target.order_reviews` r
inner join `Target.orders` o
on r.order_id = o.order_id
inner join `Target.order_items` oi
on o.order_id = oi.order_id
inner join `Target.products` p
on oi.product_id = p.product_id
group by p.product_category
order by total_reviews desc
limit 10;
```

Row	product_category	total_reviews
1	bed table bath	11137
2	HEALTH BEAUTY	9645
3	sport leisure	8640
4	Furniture Decoration	8331
5	computer accessories	7849
6	housewares	6943
7	Watches present	5950
8	telephony	4517
9	Garden tools	4329
10	automotive	4213

The most-reviewed product categories closely align with the highest-selling categories, suggesting that customer feedback activity increases with purchase volume.

Diffrent Payment Methods Used By Customers And Their Monthly Distribution

```
select extract(month from order_purchase_timestamp) as `month`,
       count(case when payment_Type='credit_card' then o.order_id else null end) as `credit_card`,
       count(case when payment_Type='debit_card' then o.order_id else null end) as `debit_card`,
       count(case when payment_Type='voucher' then o.order_id else null end) as `voucher`,
       count(case when payment_Type='UPI' then o.order_id else null end) as `UPI`,
       count(case when payment_Type != 'credit_card' and payment_Type != 'debit_card' and
       payment_Type != 'voucher' and payment_Type != 'UPI' then o.order_id else null end) as
       `not_defined`
from `Target.payments` p full join `Target.orders` o
on p.order_id=o.order_id
group by month
order by month;
```

Row	month	credit_card	debit_card	voucher	UPI	not_defined
1	1	6103	118	477	1715	0
2	2	6609	82	424	1723	0
3	3	7707	109	591	1942	0
4	4	7301	124	572	1783	0
5	5	8350	81	613	2035	0
6	6	7276	209	563	1807	0
7	7	7841	264	645	2074	0
8	8	8269	311	589	2077	2
9	9	3286	43	302	903	1
10	10	3778	54	318	1056	0
11	11	5897	70	387	1509	0
12	12	4378	64	294	1160	0

Credit cards are the dominant payment method throughout the year, while UPI consistently ranks as the second most-used option. Debit cards and vouchers account for a relatively small share of transactions, indicating a strong preference for digital and credit-based payments.

The Distribution of Payment Installments

```
select p.payment_installments ,
       count(o.order_id) order_count,
       round(count(o.order_id)*100/sum(count(o.order_id)) over(),2) as inst_percentage
from `Target.orders` o inner join `Target.payments` p
on o.order_id= p.order_id
group by 1
order by 2 desc;
```

Row	payment_installm...	order_count ▼	inst_percentage ▼
1	1	52546	50.58
2	2	12413	11.95
3	3	10461	10.07
4	4	7098	6.83
5	10	5328	5.13
6	5	5239	5.04
7	8	4268	4.11
8	6	3920	3.77
9	7	1626	1.57
10	9	644	0.62

More than half of all orders (50.58%) were paid in a single installment, while two- and three-installment plans account for another 22% of transactions, indicating that customers generally prefer shorter payment terms.

Key Insights:

1. **Customer Reach:** Target served 96,096 unique customers across 27 states and 4,119 cities, demonstrating a broad national presence.
2. **Regional Concentration:** São Paulo (SP), Rio de Janeiro (RJ), and Minas Gerais (MG) account for the highest share of customers, orders, and revenue.
3. **Business Growth:** Order volume increased by 19.76% in 2018 compared to 2017, indicating visible growth in business activity and customer demand.
4. **Seasonality:** Order activity peaked between November 2017 and March 2018, highlighting a period of heightened demand.
5. **Customer Behavior:** Most orders are placed during the afternoon and evening, making these the most active purchasing periods.
6. **Revenue Distribution:** SP generated the highest revenue by a substantial margin, establishing it as the most important market.
7. **Spending Pattern:** Average order values remain relatively consistent across states, suggesting similar spending behavior among customers.
8. **Low-Revenue States:** States with the lowest revenue often maintain relatively high average order values, indicating lower transaction volume rather than lower spending.
9. **Delivery Performance:** Delivery times vary significantly across states, with northern regions generally experiencing longer delivery periods.
10. **Delivery Estimates:** Several states receive orders much earlier than the estimated delivery date, indicating conservative delivery forecasts.
11. **Product Performance:** Home, technology, and personal care categories dominate both order volume and revenue generation.
12. **Customer Engagement:** The most purchased product categories also receive the highest number of reviews, reflecting strong customer engagement.

- 13. Payment Preference:** Credit cards are the preferred payment method, while UPI is the second most commonly used payment option.
- 14. Installment Usage:** More than half of all orders are paid in a single installment, indicating a preference for immediate payment over extended financing.

Recommendations:

1. Improve delivery performance in states with the longest delivery times.
2. Refine estimated delivery dates in regions where orders consistently arrive much earlier than promised.
3. Prioritize marketing and customer retention efforts in SP, RJ, and MG, which generate the majority of revenue.
4. Ensure adequate inventory and operational readiness during high-demand periods between November and March.
5. Continue promoting digital payment options, particularly credit cards and UPI, to enhance customer convenience.
6. Leverage high-demand categories such as Bed Table Bath, Health Beauty, and Computer Accessories for cross-selling and bundled promotions to increase the visibility and sales of related products.